Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project: Binary Sisters, Restaurant Online Ordering & Table Reservation System | Date: 6 November 2026

Run on 6 November 2026 at 22:22 (Asia/Manila), after the application click-through of Activity 3. Results come from the saved files a6_backup.txt, a6_restore.txt, a6_verify.txt, a6_access.txt and a6_recovery.txt. Items in [brackets] are for the group to fill in.

1. Cover and links

Item	Value
Our group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Owning group	Binary Sisters
Repository	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System
Our evidence folder	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests

2. Projects covered

Project	Database	Test databases	Accounts used
Binary Sisters	grp_binarysisters_projectdb	grp_binarysisters_projectdb_restore, grp_binarysisters_projectdb_recovery	rors_backup for the dump. postgres (administrator) for createdb and pg_restore, because the dump contains CREATE EXTENSION

The original database and the dump file were never modified or deleted during the recovery simulation.

3. Backup plan

Item	Plan
Objects	The whole database: 8 tables, 4 views, 7 sequences, constraints, triggers and the extensions btree_gist and pgcrypto
Method	Logical backup with pg_dump in custom format (-Fc). It allows listing and selective restore with pg_restore
Storage	A local folder outside the repository, readable only by the backup operator. Dump files are never committed. Only the name, size and checksum are recorded
Access	The rors_backup account (member of rors_backup_grp with pg_read_all_data) creates the dump. The backup folder is accessible to the DBA only
Verification	List the archive with pg_restore --list, record a SHA-256 checksum, restore into a separate database, and compare tables, row counts, constraints and a representative query
Schedule (real deployment)	Daily full dump at a quiet hour, plus a dump before each deployment

Item	Plan
Retention	7 daily and 4 weekly copies, stored on a second machine or off-site
RPO	24 hours. For an ordering system with payments, losing a day of orders is the maximum acceptable loss under this plan
RTO	1 hour to restore the latest dump and reconnect the application
Roles and privileges	Roles are cluster-wide and are not stored in a single-database dump. Table, sequence and view privileges are stored in the dump. The database-level CONNECT privilege is not restored and must be re-applied. A real deployment would also keep a roles-only dump (pg_dumpall --roles-only)

4. Backup creation and inspection

Command (password supplied through the session environment, not on the command line):

```
pg_dump -U rors_backup -h localhost -Fc -d grp_binarysisters_projectdb -f grp_binarysisters_projectdb_activity6.dump
```

Item	Value
Date and time	6 November 2026, 22:22:00
File name	grp_binarysisters_projectdb_activity6.dump
Size	51,105 bytes (about 50 KB)
SHA-256 checksum	EED5CF8012B38A31E60C2BB1109BE5301C1F63BED2234F1552F4AD4A5A807DBD
Errors	None. pg_dump exit code 0

Archive listing (pg_restore --list): the archive is in custom format, gzip compressed, with 112 TOC entries, dumped from PostgreSQL 18.6 by pg_dump 18.6. The first 60 lines captured list the extensions btree_gist and pgcrypto, the three trigger functions, the

eight tables with their sequences, and their ACL (privilege) entries. The archive holds 8 TABLE DATA entries, one per table. A file on disk is not proof of recovery, because it could be incomplete or unrestorable. Only a successful restore proves it.

5. Restoration and verification

Commands:

```
createdb -U postgres -h localhost grp_binarysisters_projectdb_restore
pg_restore -U postgres -h localhost -d grp_binarysisters_projectdb_restore
grp_binarysisters_projectdb_activity6.dump
```

Restore result: success. pg_restore exit code 0 in 0.31 seconds, with no warnings or errors. The earlier copies of the test databases were dropped first with DROP DATABASE.

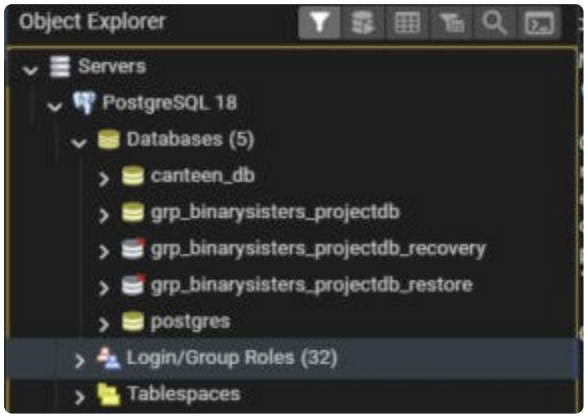


Figure 1: pgAdmin Object Explorer listing grp_binarysisters_projectdb (original) beside grp_binarysisters_projectdb_restore and grp_binarysisters_projectdb_recovery.

Check	Original	Restored	Match
Tables, views, sequences, extensions	8, 4, 7, btree_gist 1.8, pgcrypto 1.4, plpgsql 1.0	8, 4, 7, btree_gist 1.8, pgcrypto 1.4, plpgsql 1.0	Yes
Row counts: customer, menu_item, orders, order_item, payment, reservation	51, 51, 62, 171, 58, 53	51, 51, 62, 171, 58, 53	Yes
Primary keys, foreign keys, CHECK constraints	8, 6, 16	8, 6, 16	Yes
Other constraints (unique, exclusion, NOT NULL, constraint triggers)	4, 1, 45, 2	4, 1, 45, 2	Yes
Exclusion constraint ex_reservation_no_overlap	present	present	Yes

Check	Original	Restored	Match
Triggers: trg_order_item_keep_one, trg_order_item_total, trg_orders_require_items, trg_reservation_capacity	4 present	4 present	Yes
Table grants in information_schema.role_table_grants (public)	207	207	Yes
Representative query: SELECT * FROM v_order_summary ORDER BY order_id LIMIT 3	Orders 1, 2, 3: Ines Kumar 710.00, Mia Hughes 1000.00, Jack Singh 730.00	Identical three rows	Yes
Application can connect to the restored database	not required	not tested	n/a

6. Access control after restore

Table, sequence and view privileges came back with the dump. The database-level CONNECT setting did not, so PUBLIC could still connect to the new database until this was applied:

```
REVOKE ALL ON DATABASE grp_binarysisters_projectdb_restore FROM PUBLIC;
GRANT CONNECT ON DATABASE grp_binarysisters_projectdb_restore TO
rors_runtime_grp, rors_readonly_grp, rors_maint_grp, rors_backup_grp,
rors_admin_grp;
```

Test	Account	Expected	Actual
Database ACL before the fix	postgres	Restored database has no ACL	datacl was empty for the restore database, so PUBLIC could connect. The original had six explicit entries
SELECT count(*) FROM orders	rors_report	Allowed	Allowed, 62
SELECT password FROM customer	rors_report	Denied	Denied: permission denied for table customer
DELETE FROM payment WHERE false	rors_app	Denied	Denied: permission denied for table payment

Test	Account	Expected	Actual
Database ACL after the fix	postgres	Same as original	Same entries as the original database

The roles already existed because the restore went into the same server. On a new server they would have to be recreated first from the role scripts, or from a roles-only dump.

7. Recovery simulation

Only the restored copy was used.

Step	Action	Result
Before	SELECT payment_id, order_id, amount FROM payment ORDER BY payment_id LIMIT 3	Payments 1 (order 1, 710.00, gcash, paid), 2 (order 2, 1000.00, gcash, paid), 3 (order 3, 730.00, gcash, paid)
Incident	DELETE the payment row with the lowest payment_id from the restored copy. Scope: one row in a table nothing else references	DELETE 1 (payment_id 1)
After loss	The same SELECT, and a count of payment_id 1	First rows are now 2, 3 and 4. Count of payment_id 1 is 0
Recovery	createdb ..._recovery, then pg_restore of the same dump into it	Success, pg_restore exit code 0
Compare	SELECT the same row in the recovery database	Payment 1 is back: order 1, gcash, 710.00, paid. The recovery database holds 58 payments, the same as the original
Original untouched	Count of payment_id 1 in the original database	1. The original database was not affected

Outcome: Recovered. The lost payment row came back with the same values. A full restore into a fresh database was chosen because it leaves the damaged copy untouched for comparison and needs no special table-level procedure. If the loss had happened after the most recent backup, anything written since that backup, such as new orders and payments, would be lost. That window is the RPO.

8. Evaluation and recommendations

- Daily logical dumps are sufficient for a small system that can accept losing up to a day of orders. A restaurant taking payments all day would want a shorter RPO, for example continuous WAL archiving for point-in-time recovery.
- Restore time grows with data size. The test restore of this small database took 0.31 seconds, far inside the 1-hour RTO, but it should be re-measured as data grows.
- Keep a roles-only dump and the grant scripts with each backup, and re-apply the database-level CONNECT setting after every restore.
- Copy dumps off the machine and test the restore regularly. A backup that was never restored is not proven.

9. Limitations and errors

- Logical dumps capture one point in time and cannot recover to an arbitrary moment.
- The restore used the administrator account because the dump creates extensions. All restored objects are therefore owned by postgres, the same as in the dump.
- No errors or warnings occurred. The one difference after restore was the missing database-level privileges, which were re-applied as shown in section 6.
- The application was not pointed at the restored database, so application connectivity is untested. The test databases stay in place until the evidence is captured and are then dropped.

10. Feedback sent to owners

Sent on 6 November 2026: the dump and restore worked without warnings and all objects and row counts matched. The database-level CONNECT privilege is not restored and must be re-applied. The password script 03_set_passwords.psql is missing from the repository.

11. AI usage declaration

- **Tool:** Claude (Anthropic).
- **Tasks:** interpreting the results of the testing.
- **Validation:** the commands were run by the group on its own server, and every result in this report comes from the saved output files.
- **Correction made:** the AI first expected 8 sequences and 3 triggers. The restored database showed 7 sequences and 4 triggers, and the report uses the observed counts.